

ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Computer Science and Engineering

PROJECT SYNOPSIS

B.Tech 3rd Year Mini Project

HR Pulse – Enterprise Human Resource Management System

Project Title	HR Pulse – Enterprise Human Resource Management System
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College Name	ABES Engineering College, Ghaziabad
Project Guide	MR. FAIZAN
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1. Project Title

HR Pulse – Enterprise Human Resource Management System (HR Pulse)

2. Introduction

Human Resource Management is a core function of every organization. It covers employee records, daily attendance, leave handling, salary processing, hiring and performance reviews. In many small and medium organizations these tasks are still handled through spreadsheets, paper registers and scattered messages, which makes the data hard to search, easy to lose and slow to update.

HR Pulse is a full-stack web-based Human Resource Management System (HRMS) built with the MERN stack (MongoDB, Express.js, React.js and Node.js). It brings the main HR activities into one secure application with separate views for the HR Administrator and for employees.

3. Problem Statement

Manual and spreadsheet-based HR management is time-consuming and error-prone. Attendance is recorded on paper or in separate sheets, leave requests move through informal channels without a clear approval trail, payroll is calculated by hand each month, and recruitment and appraisal records are kept in different places. As a result HR staff spend a lot of time on repetitive work, employees have no direct view of their own attendance, leave balance or payslips, and management has no quick summary of the workforce.

4. Objectives

- To develop a simple, user-friendly web application for core HR operations.
- To automate attendance, leave approval and monthly payroll generation.
- To reduce manual effort and calculation errors.
- To store employee, department and HR data in one central database with role-based access.
- To give employees a self-service portal for attendance, leave, payslips and performance reviews.
- To provide dashboards and charts that summarize workforce data for HR decisions.

5. Proposed Solution

HR Pulse provides a role-based web portal. Users log in with their email and password and receive a JWT token. The HR Administrator manages employees, departments, attendance corrections, leave approvals, payroll, job openings and performance reviews. An employee uses a personal portal to punch in and out, apply for leave, view leave balance, download or print payslips and acknowledge performance reviews.

Target users: HR / Admin staff and employees of a small or medium organization.

Expected benefits: less paperwork, faster leave and payroll processing, a clear approval trail, instant in-app notifications and a single place to see workforce statistics.

6. Scope of the Project

Included: login with two roles (HR/Admin and Employee); employee directory with search, filters and CSV export; department management with budget and head; daily punch-in/punch-out with late detection; leave application and approval for Casual, Sick and Earned leave; salary breakdown and one-click bulk payroll with printable payslip; job requisitions and a candidate pipeline; performance reviews and goal tracking; in-app notifications; dashboard and report charts; dark and light theme.

Outside the current scope: mobile application, e-mail/SMS notifications, biometric or geo-fenced attendance, statutory tax filing and integration with external payroll or banking systems.

7. Methodology / Working

INPUT → PROCESSING → DATABASE → OUTPUT

- **Input:** user credentials, employee and department details, punch-in/out actions, leave requests, payroll month, job and candidate details, review ratings.
- **Processing:** the React frontend sends REST requests (Axios) to the Express API. The API verifies the JWT, checks the user's role, validates the data and applies business rules such as work-hours calculation, late detection, leave-balance update and salary computation.
- **Database:** Mongoose models store the data in MongoDB collections. A built-in in-memory fallback with automatic sample data allows the project to run without extra setup.
- **Output:** dashboards, tables, charts, printable payslips, CSV export and in-app notifications.

8. Technologies Used

Category	Technology
Programming Language	JavaScript (ES6+), JSX
Frontend	React 18, Vite, Tailwind CSS, React Router v6, Axios, Recharts, Lucide Icons
Backend	Node.js, Express.js, JSON Web Tokens (JWT), Bcrypt.js, Morgan
Database	MongoDB with Mongoose (local / Atlas, plus in-memory fallback)
Machine Learning	Not applicable
IDE	Visual Studio Code
Version Control / Hosting	Git and GitHub; frontend deployed on Vercel

9. Modules

- **Module 1 – Authentication & User Management:** login, JWT session, profile and password update, role-based access.
- **Module 2 – Employee & Department Management:** employee CRUD, 360-degree profile, search/filter, CSV export, department headcount and budget.
- **Module 3 – Attendance & Leave:** punch in/out, work hours, late detection, manual correction, leave application, balances and approval.
- **Module 4 – Payroll:** salary components, bulk payroll generation and printable payslip.

- **Module 5 – Recruitment & Performance:** job openings, Kanban candidate pipeline, star ratings, goals and employee acknowledgment.
- **Module 6 – Dashboard, Reports & Notifications:** KPIs, charts and an in-app notification centre.

10. Expected Outcome

A working, deployed web application in which HR staff can run day-to-day HR operations from one screen and employees can see their own records at any time. Compared with the manual process, the system is expected to save time on attendance, leave and payroll work, reduce calculation mistakes, keep a clear record of approvals and give management an instant view of workforce statistics.

11. Future Scope

- Mobile application for employees.
- E-mail / SMS notifications and automated reminders.
- Advanced analytics such as attrition and attendance trends.
- Geo-fenced or biometric attendance and integration with external payroll/banking APIs.
- Stronger security (refresh tokens, two-factor login) and cloud scaling.

12. References

- MongoDB Documentation – <https://www.mongodb.com/docs/>
- Express.js Documentation – <https://expressjs.com/>
- React Documentation – <https://react.dev/>
- Node.js Documentation – <https://nodejs.org/en/docs>
- Mongoose Documentation – <https://mongoosejs.com/docs/>
- Tailwind CSS and Recharts official documentation
- Project repository – <https://github.com/Ayushukla7/HRMS>
- Live application – <https://hrms-seven-delta.vercel.app/>